Co-optimization of Rooftop Solar and Storage under NEM 3.0

Model Specification (Step 9b)

Ana Santasheva

July 16, 2026

1 Overview

Step 9b is a linear program that jointly chooses the rooftop solar array size, the battery energy and power capacity, and the hour-by-hour dispatch of solar, battery, and grid, so as to minimize a household's total annual cost. Total annual cost is the sum of annualized capital cost and the net annual energy bill (imports priced at the retail time-of-use rate, exports credited at the NEM 3.0 Avoided Cost Calculator rate). The program is solved at full 8760-hour resolution, or on a reduced set of 288 representative time slices (12 months $\times$ 24 hours) weighted by days per month. All variables are continuous; there are no integer variables.

2 Nomenclature

Sets and indices

Symbol	Meaning
h	index of an hour (or representative time slice)
$\mathcal{H}$	set of hours in the analysis year: 8760, or 288 representative slices

Decision variables (all ≥ 0, continuous)

Symbol	Meaning	Units
PV	installed solar (photovoltaic) capacity	kW
B_E	battery usable energy capacity	kWh
B_P	battery power (inverter) capacity	kW
$x_h^{pv \to l}$	solar energy sent to serve load in hour h	kWh
$x_h^{pv \to b}$	solar energy sent to charge the battery	kWh
$x_h^{pv \to g}$	solar energy exported to the grid	kWh
$x_h^{b \to l}$	battery energy discharged to serve load	kWh
$x_h^{b \to g}$	battery energy exported to the grid	kWh
$x_h^{g \to l}$	grid energy imported to serve load	kWh
$x_h^{g \to b}$	grid energy imported to charge the battery	kWh
soc_h	battery state of charge at the start of hour h	kWh

Parameters

Symbol	Meaning	Units
L_h	household electricity load in hour h	kWh
G_h	solar generation per kW installed in hour h (AC yield)	kWh/kW
p_h^{imp}	retail import price in hour h (time-of-use)	\$/kWh
p_h^{exp}	export credit in hour h (NEM 3.0 ACC)	\$/kWh
w_h	weight of hour/slice h (days in month for 288 mode; 1 for 8760)	—
c_{pv}	solar installed capital cost	\$/kW
c_b	battery energy installed capital cost	\$/kWh
c_{bp}	battery power installed capital cost (default 0)	\$/kW
c_{deg}	battery degradation cost per kWh discharged (default 0)	\$/kWh
r	real discount rate (default 0.07)	—
N	analysis horizon and solar service life (25)	yr
n_{batt}	battery service life (15)	yr
$\eta_{ch},\ \eta_{dis}$	battery charge / discharge efficiency, each $\sqrt{0.96}$	—
$\underline{s},\ \overline{s}$	minimum / maximum state-of-charge fraction (0.20, 0.90)	—
$\alpha_{pv},\ \alpha_b$	solar / battery capital-annualization factors (see §4)	1/yr

3 Optimization problem

Objective. Minimize total annual cost: annualized capital cost, plus the net annual energy bill (imports minus export credits), plus battery degradation cost.

$$\begin{aligned} \min\ & PV\, c_{pv}\, \alpha_{pv} \ +\ B_E\, c_b\, \alpha_b \ +\ B_P\, c_{bp}\, \alpha_b \\ & + \sum_{h\in\mathcal{H}} w_h \Big[\big(x_h^{g\to l} + x_h^{g\to b}\big)\, p_h^{imp} \ -\ \big(x_h^{pv\to g} + x_h^{b\to g}\big)\, p_h^{exp} \Big] \\ & + \sum_{h\in\mathcal{H}} w_h\, c_{deg} \big(x_h^{b\to l} + x_h^{b\to g}\big) \end{aligned} \tag{1}$$

Constraints. For every hour $h \in \mathcal{H}$:

$$x_h^{pv\to l} + x_h^{pv\to b} + x_h^{pv\to g} \ \le\ PV \cdot G_h \qquad \text{(solar availability)} \tag{2}$$

$$x_h^{pv\to l} + x_h^{b\to l} + x_h^{g\to l} \ =\ L_h \qquad \text{(load balance)} \tag{3}$$

$$x_h^{pv\to b} + x_h^{g\to b} \ \le\ B_P \qquad \text{(charge power)} \tag{4}$$

$$x_h^{b\to l} + x_h^{b\to g} \ \le\ B_P \qquad \text{(discharge power)} \tag{5}$$

$$soc_{h+1} \ =\ soc_h + \eta_{ch}\big(x_h^{pv\to b} + x_h^{g\to b}\big) - \tfrac{1}{\eta_{dis}}\big(x_h^{b\to l} + x_h^{b\to g}\big) \qquad \text{(SOC dynamics)} \tag{6}$$

$$\underline{s}\, B_E \ \le\ soc_h \ \le\ \overline{s}\, B_E \qquad \text{(SOC bounds)} \tag{7}$$

The state of charge is cyclic: *soc* returns to its starting value over the full year in 8760 mode, or over each representative day in 288 mode. By default grid charging is disabled ($x_h^{g\to b} = 0$) and battery-to-grid export is permitted.

4 Coefficients and conventions

The capital-annualization factors convert an up-front capital cost into an equivalent annual cost over the analysis horizon N:

$$\alpha_{pv} = \mathrm{CRF}(r, N), \qquad \alpha_b = \frac{K_{batt}}{\mathrm{PVA}(r, N)}, \tag{8}$$

where

$$\mathrm{CRF}(r, n) = \frac{r(1+r)^n}{(1+r)^n - 1}, \qquad \mathrm{PVA}(r, N) = \frac{1-(1+r)^{-N}}{r}, \qquad K_{batt} = 1 + (1+r)^{-n_{batt}}. \tag{9}$$

Solar is amortized once over its N-year life. The battery term K_{batt} is the present value of *two* purchases: one at installation and a replacement at year n_{batt} within the N-year horizon. Because of that replacement, $\alpha_b > \mathrm{CRF}(r, n_{batt})$.

Prices and rates. p_h^{exp} is the NEM 3.0 Avoided Cost Calculator export credit, supplied as a month $\times$ hour table per utility. p_h^{imp} is the retail time-of-use import price for the county's utility and rate plan.

Defaults. $r = 7\%$, $N = 25$ yr, $n_{batt} = 15$ yr, round-trip efficiency $\eta_{ch}\eta_{dis} = 0.96$, state of charge confined to 20%–90% of B_E, and $c_{bp} = c_{deg} = 0$. The 15-year battery life and battery cost follow NREL ATB 2024 Residential Battery Storage; solar cost follows the CPUC-adopted installed price.

Scope of the objective bill. The bill inside the objective is a simplified hourly accounting (imports at p_h^{imp}, exports at p_h^{exp}) used to drive the sizing and dispatch decision. The full NEM 3.0 monthly accounting (non-bypassable charges, fixed charges, minimum bill, carry-forward credits, and annual true-up) is applied downstream (Step 12) when computing the reported bill.